Final assessment 2025-26

EC3040 Economics and Society

Part 1 (15 marks)

We start our analysis of the VAT on private schools with a broad overview of the costs and benefits of the policy using concepts we have learned in the course. Such a summary can easily be generated from a generative AI software. However, it is important to cross-check that the AI is not hallucinating, which is a common issue, and to edit the AI's output so that it has your own writing tone and emphasises the points that you, as an economist, think are important.

1. Using a publicly available generative AI platform of your choice (e.g. chatGPT, Claude, Gemini), generate a 300 word summary of the VAT policy. The summary should cover the main changes introduced by the policy and its likely costs and benefits. Report both the prompt that you used and the output that you received from the platform.
2. Using the course material and the IFS report provided, comment on whether the output generated seems accurate. You should go over the main claims in the output and find specific quotes from the report or the course material that support, qualify, or contradict the claims. Indicate the relevant page numbers (from the article or the lecture slides).
3. Edit the output to correct any mistakes or inaccuracies that you found, support it with references from the article or lecture notes, and rewrite it in your own tone.

Note: your answer should contain four parts: (1) the prompt you submitted, (2) the output from the generative AI platform, (3) your comments on the output, (4) the re-written summary.

Answer:

Students should provide a prompt which correctly asks the AI to provide a 300-word summary of the introduction of VAT on private school fees and its likely positive and negative effects. Prompts that do not specifically refer to the VAT on private school fees or that ask the AI to summarise the IFS report do not address the task correctly.

In the comments on the AI output, students are expected to ensure that some key features of the policy are accurately covered, including:

- *How it changes the existing policies.*

- *Do the likely positive effects mentioned by the AI include higher fiscal revenue and higher social mobility? Are these positive effects framed in terms of the positive externalities of education studied in Lecture 9, the equity argument discussed in lecture 1 and the role of redistribution discussed in lecture 2. Is the forecasted magnitude of the positive effects in line with the figures mentioned in the IFS report?*
- *Do the likely negative effects mentioned by the AI include large class sizes in public schools? Are these negative effects framed in terms of the tax distortions discussed in lecture 3? Is the forecasted magnitude of the negative effects in line with the figures mentioned in the IFS report?*

Marks should be allocated based on the student's ability to (1) identify what is a controversial claim in the generated output, (2) what evidence provided in the report addresses this claim, (3) which concepts in the lecture notes are relevant to evaluate these claims. References to the lecture slides and the report should be clearly identified with quotes, and page or slide numbers and the arguments provided in these sources should be carefully explained.

Part 2 (40 marks)

Following the broad overview of potential costs and benefits of the policy, we want to analyse the main objectives of the policy using a theoretical economic model. The model we use builds on the one introduced in Problem Set 9 but generalises it and incorporates the redistribution aspect that was missing from that model.

Public schools are free but private schools have tuition fees equal to t pounds a year. In addition to the tuition fees, families attending private school may need to pay VAT on these fees. Let τ be the VAT rate. There are three families who can attend these schools: a low-income family (L), a middle-income family (M), and a high-income family (H). Each family has one child.

The payoff from attending public school is the same for all families but depends on how many children attend the school. Let x be the number of families who send their child to public school. Suppose that the total amount of VAT raised from private school fees is used to fund public schools. In particular, let T be the total amount of VAT raised and assume that the payoff of attending public school is:

$$U_i = \frac{S}{x} + 5 \times \frac{T}{x}$$

where $S \geq 0$ is a positive number.

The benefit of attending private school depends on two things: (1) the type of the family, and (2) the total amount of fees, F, received by the private school. Let $y = 3 - x$ be the number of families that send their child to private school. The total amount of fees

received by the school is therefore $F = y \times t$. The families' benefits from sending their child to private school is:

- $u_L = P_L + 10\frac{\sqrt{F}}{y}$
- $u_M = P_M + 10\frac{\sqrt{F}}{y}$
- $u_H = P_H + 10\frac{\sqrt{F}}{y}$

where P_L, P_M, and P_H are positive numbers such that $0 \leq P_L < P_M < P_H$. The families' payoff from sending their child to private school is this benefit minus the tuition fees and the VAT.

1. [**5 marks**] Write out the payoff function of sending a child to private school for each family. Explain how each term in the payoff functions of the three citizens capture different aspects to consider when analysing the policy and how they are related to the costs and benefits you identified in part 1.

2. [**7.5 marks**] Derive conditions on the model's parameters such that, in equilibrium, only the low income family sends their child to public school and the other two families send their children to private school. We will refer to this outcome as **scenario 1** in subsequent questions. Give an example of some numerical values for the parameters such that these conditions are satisfied. *Note: what you should report are inequalities involving S, t, τ, P_L, P_M, and P_H. Explain carefully how you obtain these inequalities and why they result in the scenario described. Hint: you only need to check, for each family, whether the family would like to change its decision, given what the other families are doing.*

3. [**7.5 marks**] Derive conditions on the model's parameters such that, in equilibrium, both the low and middle-income families send their children to public school and the high-income family sends their child to private school. We will refer to this outcome as **scenario 2** in subsequent questions. Give an example of some numerical values for the parameters such that these conditions are satisfied. *Note: what you should report are inequalities involving S, t, τ, P_L, P_M, and P_H. Explain carefully how you obtain these inequalities and why they result in the scenario described. Hint: you only need to check, for each family, whether the family would like to change its decision, given what the other families are doing.*

4. [**5 marks**] Explain how changing each of the parameters (i.e. increasing or decreasing them) makes scenario 1 (only one family in public school vs. two in private school) more likely than scenario 2 (only one family in private school vs. two in public school).

5. **[5 marks]** Show carefully under which conditions an increase in VAT can increase social welfare (the sum of utilities of all the families) and under which conditions it can decrease social welfare. Show how the utility of each family changes with an increase in VAT.

6. **[10 marks]** Discuss carefully, in your own words, whether you think the introduction of VAT on private schools is a good idea. Use the results from the analysis of the theoretical model to illustrate your argument and discuss the intuition behind these results. Pay particular attention to discussing which groups would benefit or lose more from the introduction of the VAT. Make references to concept studied in the lecture wherever relevant. [Suggested word count: 300 words]

Answer:

1. *The payoff functions from private school are equal to the benefit term ($P_i + 10\frac{\sqrt{yt}}{y}$) minus the cost term ($t(1+\tau)$):*

   - $V_L(t,\tau,y) = P_L + 10\frac{\sqrt{yt}}{y} - t(1+\tau)$
   - $V_M(t,\tau,y) = P_M + 10\frac{\sqrt{yt}}{y} - t(1+\tau)$
   - $V_H(t,\tau,y) = P_H + 10\frac{\sqrt{yt}}{y} - t(1+\tau)$

   Each payoff function captures

   $$V_i(t,\tau,y) = \underbrace{P_i}_{\textit{Preference for private school}} + \underbrace{10\frac{\sqrt{yt}}{y}}_{\textit{Quality of private school}} - \underbrace{t(1+\tau)}_{\textit{Fees inc. VAT}}$$

   The first term differs across families and therefore captures each family's preference for private schools over public schools. The high-income family puts more value on private school than the middle-income one who themselves value it more than the low-income family. This could be due to different perceptions of the quality of private schools, preferences for being around like-minded families, perceived value of building a network, or perhaps because the returns to private schools depend on the parents' income.

   The second term is the same across all families and captures how much the fees received by the private school translate into improving the school. The square root function captures diminishing marginal returns to receiving more fees. The amount is divided by the number of students since the more students there are, the less resources each student benefits from. The coefficient of 10 captures the fact that the school can turn tuition fees into something valuable to the parents. Otherwise, the parents would be better off keeping the funds.

   The third term is the fees paid by the parents plus the VAT payable on the fees.

2. *We have to calculate the payoffs from public vs. private school for each family in scenario 1 and then calculate the payoff for each family from changing their decision of which school to go to.*

   First note that the total amount of VAT is equal to the total fees (the number of children in private school times the individual fee) times the VAT rate, that is: $T = y \times t \times \tau$.

   Suppose that only the low income family goes to public school and the middle and high-income families go to private school. In this case, $x = 1$, $y = 2$, *and* $F = 2t$. *The payoff to the low income family is equal to its value from going to public school when only one child goes to public school:*

$$U_L = \frac{S}{1} + 5 \times \frac{T}{1} = S + 5 \times 2 \times t \times \tau = S + 10t\tau$$

   The payoff to the high-income family is its value of private school when two children go to private school:

$$V_H(t, \tau, 2) = P_H + 10\frac{\sqrt{2t}}{2} - t(1+\tau) = P_H + 5\sqrt{2t} - t(1+\tau)$$

   Similarly, the payoff to the middle-income family is:

$$V_M(t, \tau, 2) = P_M + 10\frac{\sqrt{2t}}{2} - t(1+\tau) = P_M + 5\sqrt{2t} - t(1+\tau)$$

   Now consider the cases in which one family changes its decision unilaterally:

   (a) *Suppose that the low-income family decides to move its child to the private school. Then there would be no children in public school and three children in private school, so* $x = 0$ *and* $y = 3$. *The low-income family's payoff from doing this is:* $V_L(t, \tau, y) = P_L + 10\frac{\sqrt{3t}}{3} - t(1+\tau)$. *Therefore, the low-income family does not change its decision as long as:*

$$S + 10t\tau \geq P_L + 10\frac{\sqrt{3t}}{3} - t(1+\tau)$$

   (b) *Suppose now that the middle-income family decides to move its child to the public school. Then there would be 2 children in public school and one child in private school, so* $x = 2$ *and* $y = 1$. *The middle-income family's payoff from doing this is:* $U_M(x) = \frac{S}{2} + 5 \times \frac{t\tau}{2}$. *Therefore, the middle-income family does not change its decision as long as:*

$$P_M + 5\sqrt{2t} - t(1+\tau) \geq \frac{S}{2} + 5 \times \frac{t\tau}{2}$$

   (c) *Suppose now that the high-income family decides to move its child to the public school. Then there would be two children in public school and one child in*

private school, so $x = 2$ and $y = 1$. The high-income family's payoff from doing this is: $U_H(x) = \frac{S}{2} + 5 \times \frac{t\tau}{2}$. Therefore, the high-income family does not change its decision as long as:

$$P_H + 5\sqrt{2t} - t(1+\tau) \geq \frac{S}{2} + 5 \times \frac{t\tau}{2}$$

Therefore, scenario 1 is an equilibrium if:

- $S + 10t\tau \geq P_L + 10\frac{\sqrt{3t}}{3} - t(1+\tau)$
- $P_M + 5\sqrt{2t} - t(1+\tau) \geq \frac{S}{2} + 5 \times \frac{t\tau}{2}$
- $P_H + 5\sqrt{2t} - t(1+\tau) \geq \frac{S}{2} + 5 \times \frac{t\tau}{2}$

Note that since $P_H > P_M$, the second condition implies the third condition, so only the first two conditions are necessary.

Taking for example $S = 100$, $P_L = 0$, $P_M = 100$, $P_H = 150$, $t = 100$, $\tau = 0$ implies that:

- *The low-income family stays in public school since: $S + 5 \times \frac{2t\tau}{1} = 100 + 0 \geq \frac{10\sqrt{100}}{\sqrt{3}} - 100 = -42.26$*
- *The middle-income family stays in private school since: $P_M + 5\sqrt{2t} - t(1+\tau) = 70.71 \geq \frac{S}{2} + \frac{5t\tau}{2} = 50$*
- *The high-income family stays in private school since: $P_H + 5\sqrt{2t} - t(1+\tau) = 120.71 \geq \frac{S}{2} + \frac{5t\tau}{2} = 50$*

3. *We have to calculate the payoffs from public vs. private school for each family in scenario 2 and then calculate the payoff for each family from changing their decision of which school to go to.*

   Suppose that both the low and middle-income family go to public school and the high-income families goes to private school. In this case, $x = 2$, $y = 1$, and $F = t$. The payoff to the low-income family is equal to its value from going to public school when two children go to public school:

$$U_L = \frac{S}{2} + \frac{5t\tau}{2}$$

   Similarly, the payoff to the middle-income family is

$$U_M = \frac{S}{2} + \frac{5t\tau}{2}$$

   The payoff to the high-income family is its value of private school when only one child goes to private school

$$V_H(t, \tau, 1) = P_H + 10\frac{\sqrt{t}}{1} - t(1+\tau)$$

   Now consider the cases in which one family changes its decision unilaterally:

(a) *Suppose that the low-income family decides to move its child to the private school. Then there would be one child in public school and two children in private school, so* $x = 1$ *and* $y = 2$*. The low-income family's payoff from doing this is:* $V_L(t, \tau, y) = P_L + 5\sqrt{2t} - t(1 + \tau)$*. Therefore, the low-income family does not change its decision as long as:*

$$\frac{S}{2} + \frac{5t\tau}{2} \geq P_L + 5\sqrt{2t} - t(1 + \tau)$$

(b) *Suppose now that the middle-income family decides to move its child to the private school. Then there would be one child in public school and two children in private school, so* $x = 1$ *and* $y = 2$*. The middle-income family's payoff from doing this is:* $V_M(t, \tau, y) = P_M + 10\frac{\sqrt{2t}}{2} - t(1 + \tau)$*. Therefore, the middle-income family does not change its decision as long as:*

$$\frac{S}{2} + \frac{5t\tau}{2} \geq P_M + 5\sqrt{2t} - t(1 + \tau)$$

(c) *Suppose now that the high-income family decides to move its child to the public school. Then there would be three children in public school and none in private school, so* $x = 3$ *and* $y = 0$*. The high-income family's payoff from doing this is:* $U_H(x) = \frac{S}{3} + 5 \times \frac{0 \times t\tau}{3}$*. Therefore, the high-income family does not change its decision as long as:*

$$P_H + 10\sqrt{t} - t(1 + \tau) \geq \frac{S}{3}$$

Therefore, scenario 2 is an equilibrium if:

- $\frac{S}{2} + \frac{5t\tau}{2} \geq P_L + 5\sqrt{2t} - t(1 + \tau)$
- $\frac{S}{2} + \frac{5t\tau}{2} \geq P_M + 5\sqrt{2t} - t(1 + \tau)$
- $P_H + 10\sqrt{t} - t(1 + \tau) \geq \frac{S}{3}$

Note that since $P_M > P_L$*, the second condition implies the first condition, so only the last two conditions are necessary.*

Taking for example $S = 100$*,* $P_L = 0$*,* $P_M = 100$*,* $P_H = 150$ *and* $t = 100$*,* $\tau = 50\%$ *implies that:*

- *The low-income family stays in public school since:* $\frac{S}{2} + 5 \times \frac{t\tau}{2} = 50 + 5 \times \frac{0.5*100}{2} = 175 \geq P_L + 5\sqrt{100} - 100 \times 1.5 = -79.29$
- *The middle-income family stays in public school since:* $\frac{S}{2} + 5 \times \frac{t\tau}{2} = 50 + 5 \times \frac{0.5*100}{2} = 175 \geq P_M + 5\sqrt{200} - 100 \times 1.5 = 20.71$
- $P_H + 10\sqrt{t} - 100 \times 1.5 = 100 \geq \frac{S}{3} = 33.33$

4. *Changes in the parameters have the following impact:*

- *S: A higher S makes the second condition in scenario 1, $P_M+5\sqrt{2t}-t(1+\tau) \geq \frac{S}{2}+5\times\frac{t\tau}{2}$, less likely to hold. It also makes the second condition in scenario 2, $\frac{S}{2}+\frac{5t\tau}{2} \geq P_M+5\sqrt{2t}-t(1+\tau)$, more likely to hold. So increasing S should make scenario 2 more likely as it can make the middle-income family more likely to go to public school. A higher S also makes the low-income family more likely to stay in public school in both scenarios (see condition 1 in both scenarios), which reinforces both, but could make both scenarios less plausible if it is so high that even the high-income family prefers public school (see condition 3 in both scenarios).*
- *t: A higher t has an ambiguous impact. For the middle-income family, for instance, it increases the right-hand side of condition 2 in scenario 1 but has a non-monotonic effect on the left-hand side of that condition. To evaluate the net impact, we can rewrite the second condition in scenario 1 as follows:*

$$P_M+5\sqrt{2t}-t(1+\tau) \geq \frac{S}{2}+5\times\frac{t\tau}{2} \Leftrightarrow P_M+5\sqrt{2t}-t\left(1+\tau\frac{5\tau}{2}\right) \geq \frac{S}{2}$$

The derivative of the left-hand side with respect to t is: $\frac{\partial(P_M+5\sqrt{2t}-t\left(1+\tau+\frac{5\tau}{2}\right))}{\partial t} = \frac{5}{\sqrt{2t}}-\left(1+\frac{7\tau}{2}\right)$.

Therefore, when $\frac{5}{\sqrt{2t}} > \left(1+\frac{7\tau}{2}\right)$, that is when t is small enough relative to τ, the left-hand side increases with t. As a result, the condition for scenario 1 is more likely to hold when t increases. Since the second condition in scenario 2 is just the reverse as the condition in scenario 1, an increase in t will also make the condition less likely to hold in this case. Therefore, for small enough t, an increase in t makes scenario 1 more likely than scenario 2.

When $\frac{5}{\sqrt{2t}} < \left(1+\frac{7\tau}{2}\right)$, an increase in t makes the left-hand side of condition 2 in scenario 1 smaller and therefore makes the condition for scenario 1 less likely to hold. Since the second condition in scenario 2 is just the reverse as the condition in scenario 1, an increase in t will make that condition more likely to hold when that is the case. Therefore, for large enough t, an increase in t makes scenario 2 more likely than scenario 1.

- *τ: A higher τ makes the second condition in scenario 1, $P_M+5\sqrt{2t}-t(1+\tau) \geq \frac{S}{2}+5\times\frac{t\tau}{2}$, less likely to hold (both because it decreases the left-hand side and because it increases the right-hand side). Conversely, it makes the second condition in scenario 2 more likely to hold. So increasing τ should make scenario 2 more likely as it can make the middle-income family more likely to switch to public school. A higher τ always reinforces L's preference for public school, but a very high τ could threaten both scenarios if it makes H prefer public school.*
- *P_L: A higher P_L makes the first condition in scenario 1, $S+10t\tau \geq P_L+10\frac{\sqrt{3t}}{3}-t(1+\tau)$, less likely to hold, so would create the possibility for a third*

scenario in which all three families go to private school but does not affect whether scenario 1 or 2 is more likely.

- P_M*: A higher* P_M *makes the second condition in scenario 1,* $P_M + 5\sqrt{2t} - t(1+\tau) \geq \frac{S}{2} + 5 \times \frac{t\tau}{2}$*, more likely to hold but the second condition in scenario 2,* $\frac{S}{2} + \frac{5t\tau}{2} \geq P_M + 5\sqrt{2t} - t(1+\tau)$*, less likely to hold. So increasing* P_M *would make scenario 1 more likely.*
- P_H*: A higher* P_H *does not affect whether scenario 1 or 2 is more likely. Decreasing* P_H *could lead to a scenario in which all families send their children to public school as it makes the third conditions in scenario 1 (*$P_H + 5\sqrt{2t} - t(1+\tau) \geq \frac{S}{2} + 5 \times \frac{t\tau}{2}$*) and 2 (*$P_H + 10\sqrt{t} - t(1+\tau) \geq \frac{S}{3}$*) less likely to hold.*

5. *The utilities of each families and the social welfare function in each scenario are as follows:*

   - ***Scenario 1:***
     - *Low-income family:* $U_L = S + 10t\tau$
     - *Middle-income family:* $V_M = P_M + \frac{10\sqrt{t}}{\sqrt{2}} - t(1+\tau)$
     - *High-income family:* $V_H = P_H + \frac{10\sqrt{t}}{\sqrt{2}} - t(1+\tau)$

     The social welfare function is

$$\begin{aligned} SW_1(\tau_1) &= S + 10t\tau + P_M + \frac{10\sqrt{t}}{\sqrt{2}} - t(1+\tau) + P_H + \frac{10\sqrt{t}}{\sqrt{2}} - t(1+\tau) \\ &= S + 10t\tau_1 + P_M + P_H + 10\sqrt{2t} - 2t(1+\tau_1) \\ &= S + 8t\tau_1 + P_M + P_H + 10\sqrt{2t} - 2t \end{aligned}$$

   - ***Scenario 2:***
     - *Low-income family:* $U_L = \frac{S}{2} + \frac{5t\tau}{2}$
     - *Middle-income family:* $U_M = \frac{S}{2} + \frac{5t\tau}{2}$
     - *High-income family:* $V_H = P_H + \frac{10\sqrt{t}}{\sqrt{1}} - t(1+\tau)$

     The social welfare function is

$$\begin{aligned} SW_2(\tau_2) &= \frac{S}{2} + 5\frac{t\tau_2}{2} + \frac{S}{2} + 5\frac{t\tau_2}{2} + P_H + \frac{10\sqrt{t}}{\sqrt{1}} - t(1+\tau_2) \\ &= S + 5t\tau_2 + P_H + 10\sqrt{t} - t(1+\tau_2) \\ &= S + 4t\tau_2 + P_H + 10\sqrt{t} - t \end{aligned}$$

   Effect f* τ *on social welfare:

   When VAT increases, the social welfare within scenario 1 increases since the social welfare function is increasing in the tax rate. This is because, for any

τ_1 such that scenario 1 is an equilibrium (i.e. for any τ_1 such that all three conditions in question 2 are satisfied), then:

$$\frac{\partial SW_1(\tau_1)}{\partial \tau_1} = 8t > 0$$

This is because the VAT is not just a transfer from the high- and middle-income families to the low-income family. Every pound transferred to the public school is multiplied by 5, to reflect the high returns to education. So any decrease in utility from higher tax for the other two families is more than offset by an increase in utility for the low-income family.

Similarly, an increase in VAT increases social welfare in scenario 2 as any decrease in utility from a higher tax for the high-income family is more than offset by the increase in utility for the low- and middle-income families. This is because, for any τ_2 such that scenario 2 is an equilibrium (i.e. for any τ_2 such that all three conditions in question 3 are satisfied), then:

$$\frac{\partial SW_2(\tau_2)}{\partial \tau_2} = 4t > 0$$

However, as shown in question 4, an increase in VAT can switch the equilibrium from scenario 1 to scenario 2. Such an increase would decrease social welfare if: $SW_2(\tau_2) < SW_1(\tau_1)$, or equivalently, if:

$$\begin{aligned} & S + 4t\tau_2 + P_H + 10\sqrt{t} - t > S + 8t\tau_1 + P_M + P_H + 10\sqrt{2t} - 2t \\ \Leftrightarrow \quad & 4t\tau_2 + t > 8t\tau_1 + P_M + 10(\sqrt{2} - 1)\sqrt{t} \end{aligned}$$

This requires the new VAT rate to be sufficiently larger than the old one, relative to the value that the middle-income family attaches to private school (P_M) and the private school's fees (t).

Finally, social welfare can decrease in τ if it leads both the middle and high-income families to switch to public as the social welfare would become $SW_3(\tau_3) = 3 \times \frac{S}{3} = S$, which is lower than in the other two scenarios if $\frac{S}{3} < S + 4t\tau_2 + P_H + 10\sqrt{t} - t$ and $S + 4t\tau_2 + P_H + 10\sqrt{t} - t < S + 8t\tau_1 + P_M + P_H + 10\sqrt{2t} - 2t$.

Effect of τ on each family's utility:

Within each scenario, the utility of the low-income family increases in the tax rate, τ, in both scenarios since $\frac{\partial U_L(\tau)}{\partial \tau} = 10t > 0$. The utility of the high-income family decreases in the tax rate in both scenarios since $\frac{\partial V_H(\tau)}{\partial \tau} = -t < 0$. The utility of the middle-income family decreases in the tax rate in the first scenario since $\frac{\partial V_M(\tau)}{\partial \tau} = -t < 0$, but increases in the second scenario since $\frac{\partial U_M(\tau)}{\partial \tau} = \frac{5t}{2} > 0$.

The utility of the low-income family can either increase or decrease when we switch from scenario 1 to scenario 2. On the one hand, there are more

children in public school (x increases) so resources need to be shared and there are fewer students in private school paying fees and therefore VAT, so VAT can fall from $2 \times t \times \tau_1$ to $t \times \tau_2$. On the other hand, there is more funding from a higher VAT. Their utility increases with a higher VAT leading to a switch in scenario if $S + 10t\tau_1 < \frac{S}{2} + \frac{5t\tau_2}{2}$. Whether the utility of the middle-income family increases or decreases when the tax rate increases to a point of making them switch from scenario 1 to scenario 2 depends on the parameters. Their utility increases if $P_M + \frac{10\sqrt{t}}{\sqrt{2}} - t(1+\tau_1) < \frac{S}{2} + \frac{5t\tau_2}{2}$.

6. *Students can express their own point of view, but should ensure that each claim they make is supported by a specific result derived from the section above, or by some arguments made in the IFS report or in the lecture. Points that should be mentioned for full mark:*

   - *The VAT policy can address equity issues if it reduces the resource gap between public and private school by increasing resources in public school (See the equity argument from Lecture 1).*
   - *However, this is conditional on the total VAT rising as a result of the policy. The total VAT revenue can actually fall in scenario 2 under some conditions as fewer children attend private school, despite the higher rate. This is the case when $t \times \tau_2 < 2 \times t \times \tau_1$. If this happens, then the VAT policy would not address equity issues. This is a similar argument as the Laffer curve we discussed in Lecture 2.*
   - *In addition, the VAT policy can create some efficiency issues. This is for two reasons. First, because the VAT might make the middle-income family switch which makes them lose the utility term P_M. This corresponds to the value of "choice" over education provider discussed in Lecture 9. Second, because it creates an externality on the low-income family through the possible overcrowding of public schools, captured by the terms $S + 5T$ in the first scenario becoming $\frac{S}{2} + 5\frac{T}{2}$ in the second scenario. This connects with the broader them of externalities discussed in lecture 4.*
   - *The introduction of VAT therefore only increases social welfare if the higher funding for public schools, T, is large enough to offset (1) the fact that the middle-income family might have preferred private schools ($P_M > S$), (2) the fact that there are now more children in public schools so that S becomes $\frac{S}{2}$ for family 1, (3) the fact that there are now fewer families paying fees and therefore VAT.*

   Note, students should write out full paragraphs and give references to specific slides or page numbers in the report and course material when mentioning them.

Part 3 (30 marks)

The next step in the analysis would be to conduct some empirical tests of the policy's impact. We now want to think about how such an analysis could be conducted.

1. **[7.5 marks]** Given the results you have derived from the model above, list some predictions that the theory makes about the relationship between variables in the model. *Note: a prediction is a statement of the kind "when variable X increases, variable Y decreases", or "when variable X increases, variable Y decreases". It can also take the form of "when variable X increases, variable Y decreases, but only if variable Z is sufficiently high"*

2. **[7.5 marks]** For each prediction from the theory, explain how each variable could be measured in the real world. Find some data sources online that could be used to measure these variables. *Note: you do not need to download the data, just provide a link to the source of data and indicate which variable or combination of variables in the dataset you would use.*

3. **[7.5 marks]** Choose one of the theory's prediction from your answer to question 1 above, and write down the regression equation that would allow you to test the prediction using the data you identified in question 2. Explain carefully each term in your equation, identify the main coefficient of interest and describe what hypothesis you would test to check if the theory is correct.

4. **[7.5 marks]** Explain carefully why the regression might suffer from endogeneity issues, using the theoretical model from Part 2 whenever relevant, and propose some solutions that would allow you to address these endogeneity concerns. Discuss whether these solutions could be implemented in practice.

Answer:

1. *The theory's predictions are the following:*

   - *As the VAT, τ increases, the number of children attending public school, x will increase, so $\tau \nearrow \Rightarrow x \nearrow$. This follows directly from looking at the second and third conditions in scenario 2: as τ increases the conditions become more likely to hold, meaning that the middle and high-income families are more likely to be in public school.*
   - *As the VAT, τ increases, the number of children attending private school, y, will decrease, so $\tau \nearrow \Rightarrow y \searrow$. This follows directly from looking at the second and third conditions in scenario 1: as τ increases the conditions become less likely to hold, meaning that the middle and high-income families are more likely to switch to public school.*
   - *As the VAT, τ increases, the total amount of VAT raised, $T = y\tau t$ will increase when τ is low enough that the third condition in scenario 2 is satisfied but decrease if τ becomes large enough that the third condition in scenario 2 is*

satisfied, since fewer children attend fee-paying private schools, so $\tau \nearrow \Rightarrow T \nearrow$ *for* $\tau < \underline{\tau}$ *and* $\tau \nearrow \Rightarrow T \searrow$ *for* $\tau \geq \underline{\tau}$ *where* $\underline{\tau}$ *satisfies condition 3 with equality.*

- *As the VAT,* τ *increases, the value of public school to families attending public schools,* $\frac{S}{x} + \frac{T}{x}$ *will initially increase but eventually decrease, so* $\tau \nearrow \Rightarrow u_L \nearrow$ *for* $\tau < \underline{\tau}$ *and* $\tau \nearrow \Rightarrow u_L \searrow$ *for* $\tau \geq \underline{\tau}$*. This follows from*
- *As fees* t *increase, the benefits of being in private school,* V_i *will increase for* $t < \underline{t}$ *and decrease for* $t \geq \underline{t}$ *where* $\underline{t}$ *solves* $\frac{5}{\sqrt{2t}} = \left(1 + \frac{7\tau}{2}\right)$ *as shown in part 2 question 4 above.*

Note: there are more predictions possible.

2. *For each prediction, the variables could be measured with the following data:*

   (a) *As the VAT,* τ *increases, the number of children attending public school,* x *will increase:*

      i. *The timing and amount of VAT increase can be obtained from the government's website, e.g.* https://www.gov.uk/government/publications/vat-on-private-school-fees/applying-vat-to-private-school-fees

      ii. *The proportion of children attending public school would be better than the total number of children, since the number of children also depends on changing fertility rates. This proportion can be obtained from government's statistics, e.g.* https://explore-education-statistics.service.gov.uk/find-statistics/school-pupils-and-their-characteristics/2024-25*. The data would have to be downloaded for each year to create a time series. Note: for full marks, students should explain which variable and which dataset from this website they would use.*

   (b) *As the VAT,* τ *increases, the number of children attending private school,* y*, will decrease:*

      i. *The timing and amount of VAT increase can be obtained from the government's website, e.g.* https://www.gov.uk/government/publications/vat-on-private-school-fees/applying-vat-to-private-school-fees

      ii. *The proportion of children attending public school would be better than the total number of children, since the number of children also depends on changing fertility rates. This proportion can be obtained from government's statistics, e.g.* https://explore-education-statistics.service.gov.uk/find-statistics/school-pupils-and-their-characteristics/2024-25*. The data would have to be downloaded for each year to create a time series. Note: for full marks, students should explain which variable and which dataset from this website they would use.*

   (c) *As the VAT,* τ *increases, the total amount of VAT raised,* $T = y\tau t$ *will increase when* τ *is low enough that the third condition in scenario 2 is satisfied but decrease if* τ *becomes large enough that the third condition in scenario 2 is satisfied.*

i. *The timing and amount of VAT increase can be obtained from the government's website, e.g.* https://www.gov.uk/government/publications/vat-on-private-school-fees/applying-vat-to-private-school-fees

ii. *The amount of VAT revenue can also be obtained from government's statistics, e.g.* https://www.gov.uk/government/statistics/value-added-tax-vat- *The data would have to be downloaded for each year to create a time series. Note: for full marks, students should explain which variable and which dataset from this website they would use.*

(d) *As the VAT, τ increases, the value of public school to families attending public schools, $\frac{S}{x} + \frac{T}{x}$ will initially increase but eventually decrease.*

i. *The timing and amount of VAT increase can be obtained from the government's website, e.g.* https://www.gov.uk/government/publications/vat-on-private-school-fees/applying-vat-to-private-school-fees

ii. *The value of public school can be proxied by measures of school quality such as class sizes, offers of free school meals, or results on national exams. Some of this data is also available here:* https://explore-education-statistics.service.gov.uk/find-statistics/school-pupils-and-their-characteristics/2024-25 *while other sources also exist.*

(e) *As fees t increase, the benefits of being in private school, V_i will increase for $t < \underline{t}$ and decrease for $t \geq \underline{t}$.*

i. *The average tuition fees over time can be obtained from the ISC census:* https://cms.isc.co.uk/media/0u0llqvv/_isc_census_2025_29may25.pdf

ii. *The value of private school can be proxied by measures of school quality such as class sizes or results on national exams. Some of this data is also available here:* https://explore-education-statistics.service.gov.uk/find-statistics/school-pupils-and-their-characteristics/2024-25 *while other sources also exist.*

3. *To evaluate the impact of the VAT increase on proportion of school children attending public school, we could run the following regression:*

$$y_t = \alpha + \beta_1 \times VAT_t + \beta_2 X_t + \varepsilon_t$$

Where

- *The unit of observation t is the year of the observation.*
- *The dependent variable y_t would be the share of pupils in England attending public school vs. those attending private schools.*
- *The independent variable VAT_t is a dummy variable taking value 1 in the first year after the VAT increase, i.e., 2025.*

- *X_t is a set of control variables that could include for example some proxy for other factors affecting the decision of going to private school such as average income, relative school quality in the previous year (e.g. as measured by success on GCSEs or A-levels in public vs. private schools), etc.*

The coefficient of interest is β_1. The theory predicts that the introduction of VAT will lead to an increase in public school attendance, so our null hypothesis should be $H_0 : \beta_1 = 0$ and alternative hypothesis is $H_1 : \beta_1 > 0$. A t-test of this hypothesis can tell us whether the relationship is statistically significant.

4. *Endogeneity issues arise when the independent variable is systematically correlated with the error term. This can arise for two reasons: unobserved omitted variables which are systematically correlated with both the introduction of VAT and public school attendance and reverse causality, whereby the introduction of VAT is determined by the level of public school attendance.*

   The main concerns with this regression is due to unobserved omitted variables. Besides the control variables listed above, there could be other factors that can affect the decision to send children to private school that change over time but cannot be measured. For instance, parents' taste for private school might be changing year-on-year (captured by parameters P_M, P_H and P_L in the model) and be correlated with other political preferences (e.g. preferences for redistribution). This only creates an endogeneity issue if changes in these factors are systematically correlated with the introduction of VAT.

   For instance, this could the case here because an increase in preferences for redistribution across the population might have been what led to the election of the Labour party, who introduced the VAT. If people's preferences for redistributions are correlated with their preferences for public school over private school (i.e., a lower P_i), then an increase in preferences for redistribution would both lead to the VAT being more likely to be introduced AND an increase in attendance in public schools, EVEN IF there were no direct causal effect from the VAT increase to the attendance decrease.

   This does not mean the increase in VAT does not lead to an increase in public school attendance, but that the effect of the VAT could be overestimated, as some of the effect could be simply driven by the confounding variables. If changes in political preferences towards Labour are positively correlated with a change in preferences towards public school, then our estimate of β_1 would overestimate the true treatment effect. Without a measure of the underlying change in preferences, we cannot identify how much of the change in the proportion of children going to public school is due to the VAT or a change in taste.

   Some possible solutions would be to use time trends from previous years to net out the change over time, if there is a clear linear trend. To do this, we could for example run a difference-in-differences specification across months and local authorities where the treatment group are the months around the introduction of

the reform but in the year prior to the reform. This would address the endogeneity issue by subtracting the underlying long-term change in unobserved preferences for public school from the change in public school attendance around the time of the VAT introduction.

Alternatively, we could conduct a difference-in-differences by comparing the share of public school attendance in Northern Ireland to the share in counties in the Republic of Ireland close to the border, since the Republic of Ireland did not introduce a VAT on private schools. In this case, the treatment group would be counties in Northern Ireland and the control group would be counties in the Republic of Ireland. This would address the endogeneity issue because omitted variables such as preferences for public school are unlikely to change systematically differently across each side of the border so subtracting the difference over time in attendance in the Republic of Ireland would net out the effect of change in unobservable preferences.

Part 4 (15 marks)

Discuss how the theoretical model from Part 2, and the evidence we could get to test it (as derived in Part 3), can shed light on the various arguments brought up on the two Youtube videos posted in the documents (https://www.youtube.com/shorts/PPEbYZykG1A and https://www.youtube.com/shorts/Gh-YtzX3Hcc). What arguments are brought up in the video but not included in the model? How would you change the model to analyse these points? [Suggested word count: 300 words]

Answer: Students should make specific references to the claims made in each video and for each claim indicate clearly which element of the model helps support or contradict the claim.

For example, in the first video (The danger of VAT on school fees), the following arguments are made:

- *The VAT on private school fees will increase cost of private school and mean families who send their children there will have less disposable income to spend on other things (at 18 seconds into the video). This is a point covered in the IFS report but not covered in the model.*

  Introducing it in the model is unlikely to change the main conclusion about the changes in attendance but could change the consequences about social welfare if the lower spending power would have had a larger multiplier on GDP than the fiscal multiplier from redistributing towards public education (captured by the coefficient of 5 on T in the model).

  To introduce it in the model, we would need to model the consumption and saving decisions of families, in addition to their decision over where to send their children. For instance, we could say that the utility of a family from sending their child to

private school is equal to $V_i(c, s, x) = u_i(c) + P_i + \frac{\sqrt{(F)}}{y}$ *subject to* $c + (1+\tau)t + S \leq I$ *where* I *is the family's income and* c *is their consumption. As a result, an increase in VAT would make the budget constraint more binding and force the family to reduce* c*. To measure the effect on social welfare, we would need to introduce some parameter in the utility of all families that depends on the aggregate consumption or alternatively have a full general equilibrium model of production and consumption.*

The importance of the multiplier in determining the effect of the policy on welfare is missing from the video but highlighted in the lecture when discussing the positive externalities of education (Lecture 9) and the negative externalities of inequalities (Lecture 1). The model captures it through the coefficient of 5 on the term T *in the utility of public school.*

- *If the VAT does not lead to reduced consumption, it will lead to people pulling their children out of private schools (at 26 seconds into the video). This argument is captured by the model as we showed in Part 2 question 4 that an increase in VAT can lead to an increase in public school attendance. In particular, the term* $\frac{S}{x}$ *captures the potential effect of overcrowding that arises when* x *increases as a result of the VAT.*

  However, the video misses some important point we learnt from the model. First, the decrease in public school capacity could be outweighed by the increase in funding (T*). In fact, this is the main point made by the government and highlighted in the IFS report (as we learned in Part 1). Second, the decrease in private school attendance could affect the amount of VAT raised directly. The first point qualifies the fear that the VAT might be detrimental to public schools, as we have shown in part 2 question 5 that the net effect on social welfare could still be positive. The second point would reinforce the argument from the presenter but is not mentioned.*

- *People who would have sent their children to private school might now decide to send them to public school, further decreasing the capacity in public school. This is essentially the same argument as the previous one and is therefore also captured, but qualified, by the model.*

- *The increase in public school attendance will lead teachers striking. This point is not captured directly by the model, though we could argue that this is one of the negative effects captured by the* $\frac{S}{x}$ *term. What the model shows is that as long as the increase in funding* T *outweighs the increase in attendance* x*, then the policy has a net positive effect on social welfare. That's the case regardless of whether the effect of increased attendance goes through larger class sizes or strike from teachers. If we wanted to introduce it in the model, we could add an additional agent which represents teachers and allow them to choose between striking or not striking, where striking comes at a cost but can increase their payoff (e.g. obtain a higher wage). The decision to strike could be influenced by the number of children in public school,* x*, thus capturing the argument of the presenter, and strike would in turn be introduced as a negative effect on the public school children's utility,*

whenever teachers choose to strike. However, ultimately, the effect of overcrowding on strikes and in turn the effect of strikes on the utility from public school would be very similar to the effect captured by the term $\frac{S}{x}$ so adding this to the model would not add much.

Similar discussions could be made about the arguments made in the second video.

Note: this is an example of arguments that could be made. The word limit will necessarily constrain how much the student can discuss, but the key elements for a high grade would be for the student to discuss specific arguments made in the videos, clearly connect them to the model, and suggest specific changes to address the limitations of the model.